

Mahesh Ramesh

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EDUCATION

- **University of Wisconsin-Madison**

M.S. Computer Science GPA 3.87/4.00

Madison, WI

Sep 2024 – May 2026

- **Indian Institute of Technology Madras**

B.Tech Chemical Engineering CGPA 8.64/10

Chennai, India

Jul 2019 – Jul 2023

SKILLS AND KEY COURSEWORK

- Python, C++, SQL, PyTorch
- Machine Learning, Natural Language Processing, Probability Theory, Foundation Models, Theory of Reinforcement Learning, Game Theory Optimization and Learning

PROFESSIONAL EXPERIENCE

- **Kalpa Labs (YC F25)**

Machine Learning Intern

San Francisco, USA

Feb 2026 – Present

- Extended Megatron-LM to support large-scale TTS training, including dense and MoE model variants.
- Ran architectural studies across backbone and depth-decoder designs, evaluating multiple TTS variants.

- **JP Morgan Chase & Co - Asset & Wealth Management**

Quantitative Research Analyst

Bangalore, India

Aug 2023 – Aug 2024

- Developed an outlier-detection engine for key risk sensitivities (CSO1, DVO1, Equity Delta, Commodity Delta, Vega), increasing alert precision and automating numerous manual sanity checks.
- Improved the product-risk rating framework with idiosyncratic events, enabling more accurate risk stratification.

- **WorldQuant LLC**

Quantitative Research Intern

Mumbai, India

May 2022 – Jul 2022

- Developed C++ modules to weight a large volume of alphas and merge them into meta for production portfolios.
- Implemented constrained optimization for weight assignment and applied forecasting techniques to enhance model performance.

RESEARCH EXPERIENCE

- **ICML 2026 (In Submission, Scores: 5/6, 5/6, 5/6), NeurIPS 2025 (LAW Workshop)**

Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents

- Introduced a new multi-turn benchmark to evaluate the state-tracking ability of frontier models and performed systematic ablations with context engineering, Best-of-K sampling, and multi-agent scaffolding.
- RL Trained a 4B-parameter model on our curated dataset, outperforming all existing non-reasoning baselines and showed generalization in cooperative and temporal reasoning.
- Released an open-source dataset with 1500+ game trajectories for Supervised Fine-Tuning (SFT) and move-level ratings from SOTA reasoning models for Reinforcement Learning with Verifiable Rewards (RLVR).

- **ICCV 2025 and CVPR DemoDiv Workshop (Oral)**

CuRe: Cultural Gaps in the Long Tail of Text-to-Image Models

- Curated a dataset of 300 artifacts covering 6 cultural axes, 32 categories, and 64 countries by mining the Wikimedia knowledge graph for hierarchical benchmarking
- Developed Marginal Information Attribution metrics, which achieved a 2x increase in Spearman ρ correlation with human cultural representativeness ratings versus previous proxy scorers.
- Conducted 2,700 artifact-level surveys with culturally-aligned annotators to assess similarity, representativeness, diversity, offensiveness, and stereotypes.

- **Deployable AI Workshop, AAAI 2024 (Oral)**

MABViT: Modified Attention Block Enhances Vision Transformers

- Enhanced transformer efficiency by integrating Gated Linear Units (GLUs) into the attention module, enabling better parallel computation of MLP and attention blocks (precursor to G2 in Qwen's Gated Attention).
- Achieved a 0.6% accuracy improvement over ViT-S/16 and surpassed ViT-B/16 performance with half the parameters; MABViT models also demonstrated 17% faster training convergence.
- Improved pattern recognition, outperforming traditional ViT models in deeper architectures, an enhancement for SOTA pipelines like Gemini Flash.